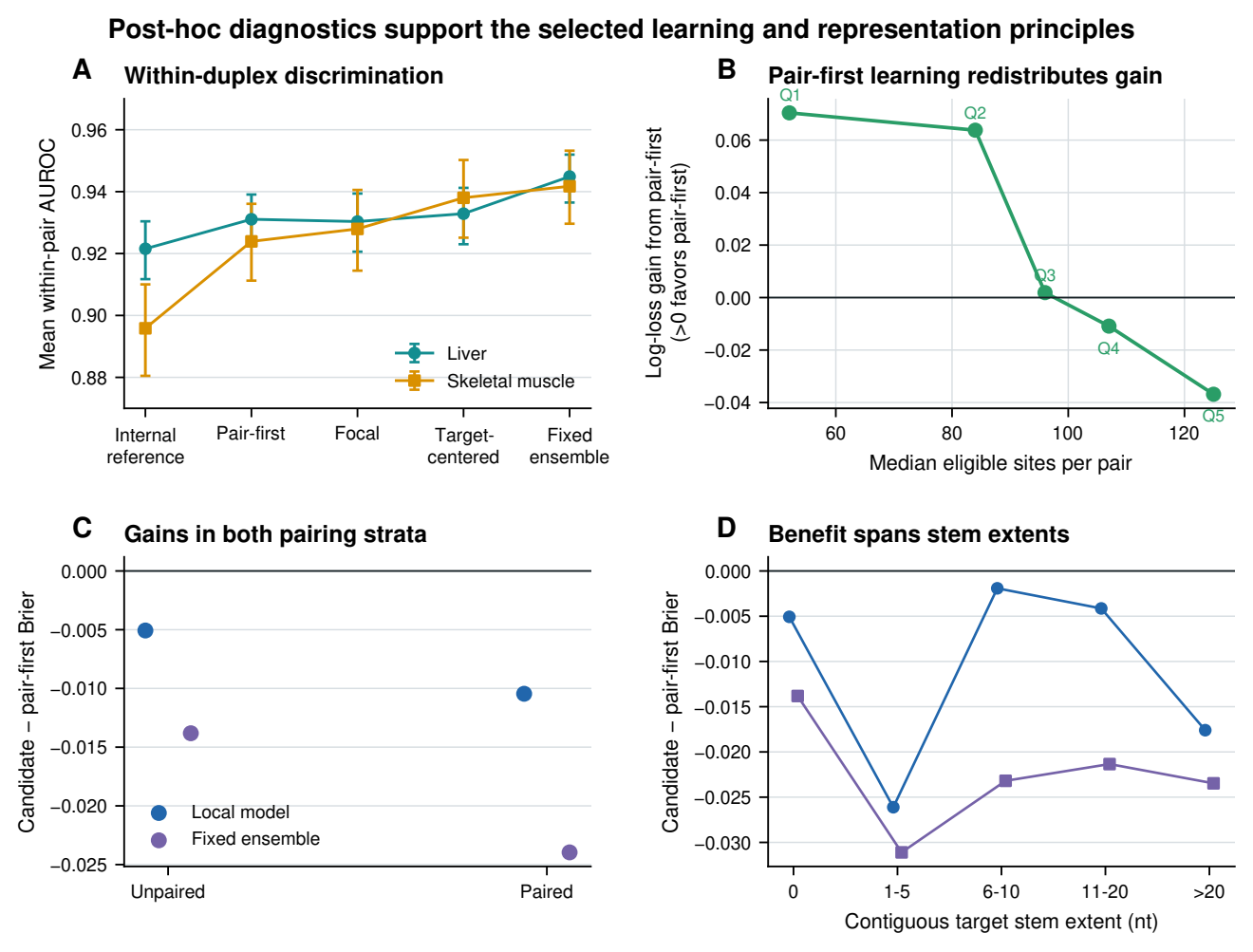Post-hoc diagnostics support the selected learning and representation principles